

Special Problem Proposal: Diagnosis Prediction Model for
Pediatric Appendicitis using Artificial Intelligence
Transformer Model

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1 Introduction

1.1 Background of the Study

Appendicitis or the inflammation of the appendix, in most healthcare institutions, is considered as one of the most common diseases that pediatric patients experience, roughly affecting 5% of the population [1, 2, 3, 4, 5]. People who often times experience abdominal pain or forms of discomfort commonly dismiss these experiences creating implications that mild symptoms do not generally equate to having appendicitis. In the instance that a person experiences mild discomfort in the lower area of the abdomen, it is common for the pain to be constant and consistent, sometimes pain fluctuates in irregular timings, and coughing may cause a sharp pain in the area [6, 7]. Due to sudden occurrences of appendicitis, medical professionals opt to use different methods to shorten the time needed to address the situation and to reach a diagnosis. From childhood to adulthood, specifically patients ranging from ages 5 to 45, appendicitis is a common problem that can be solved through numerous acts of surgery and medical intervention [2, 5]. However, due to the severity of the disease and the commonality it presides, appendicitis is often a major concern by medical professionals. Therefore, due to disregard to early-stage symptoms, the inflammation of the appendix worsens over time. According to Ansari, *"If untreated, necrosis, gangrene, and perforation occur and If the perforation is contained by the omentum, an appendiceal abscess results"* [3]. Detection of early-stage symptoms is paramount in treating acute inflammation. There are cases where children and adolescents are more prone to acquiring appendicitis. Yet, majority of medical cases involving appendicitis often appear within the children and adolescents while ages over 45 are less likely to occur [2].

Consequently, patients to be diagnosed with appendicitis will undergo physical examination, where a doctor consults the patients regarding their symptoms and a test where they apply pressure in certain abdominal areas to detect inflammations and such [8]. After consultation, the doctor instructs the patient to undergo a series of laboratory testing, such as: Blood Tests, Urine Tests, and Imaging Tests [7]. Blood and Urine tests are used to determine any bacterial infection within the bloodstream and urinary tract [7]. Imaging tests originate from Ultrasound, Computerized Tomography (CT Scan), and the Magnetic Resonance Imaging (MRI Scan) to view the abdominal area [7, 9].

To note, it is claimed by Domain Experts that amongst processes of diagnosing, False Positives (Tests imply that they have a disease while a patient does not have such) and False Negatives (Tests imply that they do not have the disease while a patient does have such) are common reoccurring problems experienced in concluding a diagnosis. Research papers emphasize the implications and risks of immediate acceptance of a certain result considering that this determines the next course of action. Similarly, in the case of diagnosing for appendicitis, it is of best practice to carefully analyze results despite its plausibility and reasoning due to a possibility of it being false [10]. While the problem of false positives and negatives are present in different fields of study, this itself proves that diagnosis do come with various factors that may hinder it from being a successful True Positive or True Negative case. Likewise, research papers introduce the idea of implementing different methods and practices to limit the presence of false positives and false negatives through the use of AI-Driven systems and models to capture plausible outcomes and reduce falseness based on existing data [11, 12].

In the Philippines, local hospitals utilize two (2) different scoring tools (Alvarado and RIPASA) to easily predict diagnosis for appendicitis. The common criteria of the Alvarado Scoring used is MATRELS: Migration of Pain, Anorexia, Nausea, Tenderness in right lower quadrant, Rebound pain, Elevated temperature, Leucocytosis, Shift of white blood cell count to the left [13]. On the other hand, RIPASA scoring criteria covers more parameters than Alvarado, some notable are: Sex, Age, RIF pain, Fevers, Duration of Symptoms, etc [14, 15, 16, 17]. Although different institutions practice using either tool, it questions the accuracy of the prediction [18]. Consequently, despite the questionable accuracy brought upon by these scoring metrics, it is primarily used as a screening tool for medical professions to gauge the possibility of appendicitis. It is worth mentioning that solely using a scoring metric in the field of medicine, often questioned for its accuracy, is impractical since reaching a diagnosis is important in providing the correct medical attention and services to each patient. There are studies that circulate comparing which tools provide the most accurate outcome, yet despite such, healthcare professionals prefer waiting for the aforementioned laboratory results before diagnosis whilst using Alvarado and RIPASA to initially screen for a more plausible outcome.

Artificial Intelligence or commonly known as AI, is the application of using programming and technology to build a virtual version of a human mind [19, 20]. Often

described as a systematic approach to build synthetic minds that can replicate human-related interactions. Considering that AI is a virtual and synthetic mind that replicates human-related actions, it disregards certain factors like morals and ethics which is an obstacle that needs to overcome, especially in the field of medicine [21, 22]. Therefore, human intervention is often needed and required when operating AI-related systems and models. Nonetheless, artificial intelligence is continuously being used and practised for improving healthcare systems. Since AI has the ability to understand complex datasets and information, medical institutions use these systems to process data in large quantities. Processing data is paramount to simplifying medical records and to understand certain diagnostic patterns and information [23, 24, 25].

Subsequently, research studies on applications of artificial intelligence, machine learning, and deep learning are done regarding appendicitis detection. Likewise, pre-existing and on-going studies emphasize the further need of conducting such studies to improve methods of utilizing such in the medical field. Within these studies, algorithms such as decision trees, vector support machine, etc. are often used to detect relationships through sorting and branching-out patterns where results show that older algorithms are still efficient in creating a diagnosis prediction model [26, 27, 28, 29]. For various elements under specific fields of data, it is difficult to sort such if only a singular algorithm is used, thus paper mentioned that detecting relationship between correlated elements require different algorithms to fully encapsulate key relationships. Furthermore, transformer models are gradually being used in medical-related practices due to its ability to capture and highlight relationships between correlated variables and elements.

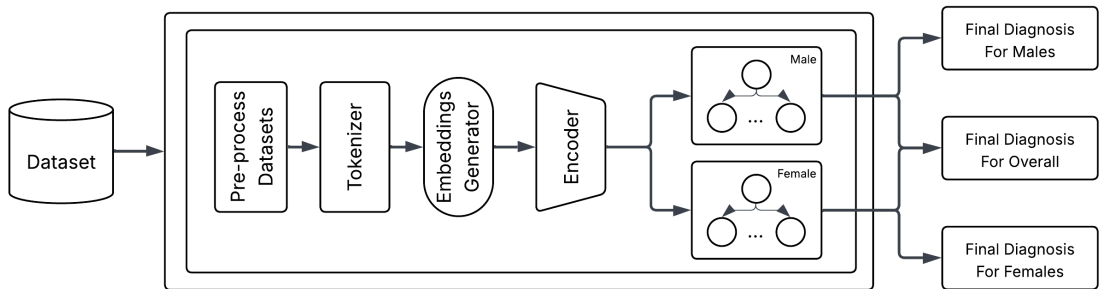


Figure 1. Transformer Model for Diagnosis Prediction

Understanding the correlation between determining factors in appendicitis is crucial for early-stage diagnosis. The transformer model is an artificial intelligence model that specializes in relationship analysis, specifically in understanding the relationship between

one variable to another [30]. More so, Figure 1, is a simplified process diagram which simulates how data is handled in the case that the Transformer Model is applied in this specific study. This diagram emphasizes that essential data should undergo pre-processing techniques to eliminate inconsistencies and skewness between outputs [31]. Furthermore, it is important to note that separate processes will be implemented for both males and females to eliminate very skewed outcomes, especially that chemical compositions within genders vary in levels [32].

Therefore, applying artificial intelligence, specifically transformer models as a decision making system, can improve current methods to determine appendicitis cases. Accompanied with medical datasets that are preprocessed, the decision system can use datasets to analyze and compare core relationships and features in producing a plausible diagnosis for appendicitis.

1.2 Statement of the Problem

Diagnostic scoring metrics for appendicitis, such as Alvarado and RIPASA, are commonly used in initial assessments but are often insufficient when used alone, as majority of medical professionals rely on additional supporting evidences such as laboratory results before concluding a diagnosis. While existing studies emphasize the advantages of artificial intelligence in appendicitis prediction, many of these studies utilize older machine learning algorithms and a combination of various models to correlate different groups of variables and factors. Although these approaches allow each algorithm to detect and highlight relationships between features, trying to replicate such practice is difficult due to its resource-intensive requirements to run such. Moreover, while there increase practices and adoptions of advanced artificial intelligence in various medical domains, no current studies specifically explore the use of transformer models to appendicitis prediction and diagnosis. This unexplored area of study limits the opportunity to analyze and evaluate whether the transformer models could simplify the prediction process by replacing and/or reducing multiple algorithms with one model.

1.3 Objectives of the Study

1.3.1 General Objectives

To develop an Appendicitis Diagnosis Prediction Model using Transformer Models while incorporating evaluation metrics and domain expert validations, to determine the efficiency of transformer models in determining key relationships between variables, and to cater the diagnosis prediction model based on new datasets.

1.3.2 Specific Objectives

1. Objectives for the Medical/Domain Expert:
 - (a) View patient data and information provided to the system, data includes medical history and patient attributes.
 - (b) Review datasets used for training and testing for analysis and evaluation
 - i. View each entry used in the system
 - ii. Add new patient data entries in the system
 - (c) Receive a prediction of a diagnosis with:
 - i. Evaluate predictions based on their medical expertise.
 - ii. Use a re-calculate option to reconsider and retry the diagnosis prediction to reach a new conclusion.
2. Objectives for the Medical Student:
 - (a) Enter patient data into the model for analysis
 - (b) Receive the AI-generated appendicitis diagnosis prediction.
 - (c) Utilize the prediction as insights for learning and case familiarity.
3. Objectives for the AI Expert:
 - (a) Dataset and Preprocessing Responsibilities
 - i. Perform all required machine learning preprocessing techniques and methods:
 - A. Handling missing entries.
 - B. Removing variables with mostly missing values.

- C. Removing duplicate entries.
 - D. Sorting or categorizing demographic data (e.g. Male and Female)
 - E. Ensuring features align with appendicitis-related attributes.
- Load new preprocessed datasets into the system.
- ii. Partition datasets into training and testing sets with a 60:40 ratio, respectively.
 - iii. Preview datasets loaded in the system, including sorting and viewing entries
- (a) Model Configuration and Evaluation
- i. Select any preprocessed and tokenized dataset loaded into the system.
 - ii. Define patient attributes and variables used as model inputs.
 - iii. Specify target to either Appendicitis or No Appendicitis
 - iv. Choose available model (Decision Trees, XGBoost, Gradient Boosting)
 - v. Evaluate model performance using medical statistics metrics (Sensitivity, Specificity, PPV, NPV)

1.4 Significance of the Project

The research aims to consider the use of transformer models in the field of diagnosis prediction, specifically for appendicitis cases. Considering traditional scoring tools and pre-existing solutions that utilize the combination of algorithms, it paves a path for a refined solution to reduce the use of multiple combined algorithms for processing specific variables while being able to fully capture any underlying relationships between correlated variables and factors [18, 26, 27]. Likewise, a transformer model that highlights the relationships can help in determining a plausible and reasonable diagnosis.

The findings of this study will be of value to Gastroenterologists by providing additional insights that may complement existing diagnostic approaches such as scoring metrics like Alvarado and RIPASA. The study will also benefit researchers in artificial intelligence and machine learning by presenting techniques and processes that can support future studies on AI-assisted diagnostic systems.

1.5 Scope and Limitations

The following are the scope and limitations present in the study and development of the diagnosis prediction model:

1. The project only uses publicly available medical datasets. Due to medical-related laws on privacy, the research will opt to use publicly available datasets found in websites such as, Kaggle and UCI Machine Learning Repository. Likewise, the datasets under St. Hedwig's Pediatric Department in Regensburg, Germany will be used in the research study, capturing and will be limited to certain aspects:
 - (a) Two (2) different datasets will be used to train the model.
 - i. The first dataset contains 782 entries and 58 different parameters/features.
The second dataset contains 624 entries and 56 different parameters/features.
 - ii. Total entries of 1406 will be partitioned as 60% training and 40% testing.
 - (b) The datasets include demographic data, clinical scores, list of symptoms, laboratory results, and imaging results.
 - (c) In the case that the dataset contains unused parameters, preprocessing techniques will be applied to eliminate the mentioned parameters.
2. The study will primarily focus on acute appendicitis rather than chronic appendicitis. In a study by Kim et al. 2016 titled, "*Chronic appendicitis in children*", emphasizes that cases of chronic appendicitis in children are rare and often misdiagnosed due to medical complications [33].
3. The evaluation of the domain expert to the usage and outputs of the diagnosis prediction model will be limited to the publicly available datasets.

1.6 Assumptions

The research study assumes the following cases:

1. The diagnosis prediction model will only cover pediatric appendicitis. Although, appendicitis may be experienced within adults aged over 30 years old, the system will primarily focus on the population ranging from ages less than 30 years old or those that are recorded to acquire appendicitis.

2. The healthcare professional will never input missing diagnosis and basic information (e.g. Symptoms, Previous Medical Records, Age, Sex, etc.).
3. The healthcare professional will never apply the diagnosis prediction model to patients older than 18 years old.
4. Used datasets from Kaggle will suffice as basis for diagnosis prediction for pediatric appendicitis.

2 Review of Related Literature

2.1 Transformer Model on Healthcare Datasets

As healthcare data is said to consume 30% of the global data ecosystem or a massive network of data, it implies that due to the rapid growth of both technological advances and the human population, the 30% mentioned is to growth at a larger scale [34, 35, 36]. Over the years, technological advancements have been used in practised to improve healthcare services and to reduce any excessive and resource-intensive processes resulting in healthcare quality [34]. The rise of artificial intelligence has paved way for improvements in the field of medicine, yet, it is without a doubt, that further improvements are still needed to address various underlying issues. In a study titled, *"Transformers and large language models in healthcare: A review"* by Nerella, S. et al. 2024, the study focused on collating various medical journals and articles into a massive database to be processed by the AI system. Considering the mentioned process regarding the transformer model, the study emphasizes that a criteria must be strictly implemented to improve the accuracy of the transformer model in the said database [34]. Given the ability of self-attention, having inconsistencies within a dataset possess a liable risk for both the system and the data used. Likewise, the preprocessed data, once fed to the system, had discovered the certain patterns or keywords that are often used in certain categorical medical studies or papers [34, 36]. This itself is a testament on how self-aware artificial intelligence can aid in detection, analysis, and prediction, not only for random sampled data, but specifically in the field of diagnosis prediction.

2.2 Research Journals on Appendicitis Diagnosis Prediction Models

2.2.1 Study 1: Decision Model for Acute Appendicitis Treatment With Decision Tree Technology — A Modification of the Alvarado Scoring System

Similar to majority of papers pertaining to creating a decision-making model for acute appendicitis, the paper titled *"Decision Model for Acute Appendicitis Treatment With Decision Tree Technology — A Modification of the Alvarado Scoring System"* emphasizes the commonness of acute appendicitis in hospitals thus highlighting the need to properly diagnose patients. However, concluding a diagnosis and decision often becomes a diffi-

culty in medical professionals especially in time-based pressure during sudden emergency situations [37]. To address such problem, the paper had the following machine learning models and techniques such as Bayesian Statistic Models, Discrimination Rules, Logistic Regression, and Neural Networks to fully construct a abdominal pain diagnostic model [37]. Alongside the aforementioned models and techniques, the paper strictly focuses on using Decision Trees as its core model in developing a decision support model for acute appendicitis diagnosis. In the development of the model, the researchers explain that utilizing the outputs of Alvarado can significantly boost the possible diagnosis provided that the scoring metric can gauge the possibility of appendicitis. Consequently, the usage of Decision Trees was deemed efficient such that utilizing varying techniques alongside Alvarado scoring metrics can provide accurate diagnostic outcomes.

2.2.2 Study 2: Light gradient boost tree classifier predictions on appendicitis with periodontal disease from biochemical and clinical parameters

The study titled, *"Light gradient boost tree classifier predictions on appendicitis with periodontal disease from biochemical and clinical parameters"* defines appendicitis as a life-threatening situation such that forms of inflammation, when not treated immediately, may cause further and extreme levels of medical complications [2, 38]. The researchers highlight that in creating systems for diagnosis, prediction, and decision-making, variables and features correlated to the disease should undergo forms of evaluation to determine the importance of each element while considering the presence of different risk factors [38]. The idea of evaluating variable and importance between features is essential in the development of a transformer model for diagnosis prediction since variable relationships are highlighted and used to detect correlation. To address the situation, the paper utilized a Light Gradient Boosted Trees (LightGBM, a variant of Gradient Boosting) due to its speed and efficiency in determining the best possible outcome. Subsequently, the application of such machine learning techniques to predictions of appendicitis had clearer visualization and insights due to the speed of tree construction, making the decision-making process faster and more efficient. However, the paper notes its limitation such that a broader and more general audience or patient data may have a small to significant output to the predictive output [38].

2.2.3 Study 3: Artificial Intelligence and Acute Appendicitis: A Systematic Review of Diagnostic and Prognostic Models

A study titled, "*Artificial Intelligence and Acute Appendicitis: A Systematic Review of Diagnostic and Prognostic Models*" by Issaiy, et al. 2023 analyzes the extent of the applications of artificial intelligence as means of diagnosis and prognosis [26]. Diagnosis can be defined as the process of identifying the disease or sickness a person has whilst Prognosis can be defined as the process of identifying the likeliness of a person to acquire that specific disease or sickness [39]. Similarly, the research paper mentions that AI application for appendicitis detection can greatly improve the time spent for diagnosis considering that laboratory results are often time-consuming and the speed up of such will allow patients to attain the medical attention needed [26]. Furthermore, the study emphasizes the methodology done wherein data gathered is selected and sorted according to a given criteria which is extracted, synthesized, and analyzed for the AI model [26]. Consequently, the research utilized a Prediction Model Risk of Bias Assessment Tool (PROBAST) to eliminate any form of biases within the dataset since any form of bias will result in a skewed outcome [26, 40]. Overall, using algorithms like Decision-Trees and K-Nearest Neighbours had concluded that using artificial intelligence as a medium in a diagnosis prediction model had showed significant outcomes where it had surpassed traditional medical methods and scoring tools.

2.2.4 Study 4: Artificial intelligence for predicting acute appendicitis: a systematic review

Likewise, a study titled, "*Artificial intelligence for predicting acute appendicitis: a systematic review*" by Lam, et al. 2023 specifically tackles diagnosis prediction models for acute appendicitis within pediatric patients [27]. In the study, there shows a table containing all of the models and algorithms used alongside the parameters taken for each model. A list of algorithms and models used in the development of the AI model are: Support Vector Machine, Artificial Neural Networks, Fuzzy Logic, Voter Algorithm, Supervised Learning Algorithm, Limited Memory, BFGS Algorithm, Naive Bayes Algorithm, Generalized Linear Model, Random Forest Algorithm, Kth Nearest Neighbour Algorithm, Decision Tree Algorithm, Support Vector Machine, Random Forest, Classification and Regression Tree, Logistic Regression, Gradient Boosting Machines, and

Binary Logistic Regression [27]. Do note that each model used in this research study utilizes different parameters based on specific objectives or goals for the study [27]. Similarly, the research study utilized the same data processing techniques as the research by Issaiy, et al. 2023, where PROBAST was used for data pre-processing and analyzes [26, 27]. Nonetheless, the research provided positive results concluding the research that AI applications will improve diagnosis of appendicitis in healthcare systems [27].

2.2.5 Study 5 and 6: Predicting severity of acute appendicitis with machine learning methods: a simple and promising approach for clinicians — Machine-learning-assisted Preoperative Prediction of Pediatric Appendicitis Severity

In addition, Machine Learning (ML) is a field inside the concepts of artificial intelligence (AI) as ML focuses on pattern recognition through data training towards a model [41]. Studies titled, *"Predicting severity of acute appendicitis with machine learning methods: a simple and promising approach for clinicians"* by Yazici, et al. 2024. and *"Machine-learning-assisted Preoperative Prediction of Pediatric Appendicitis Severity"* by Erman, et al. 2025. are research papers that pertain to specific uses of machine learning algorithms to detect pediatric appendicitis [28, 29]. Both researchers approached the issue of diagnosis prediction model by analysing data based on various parameters similar to the research mentioned earlier. Furthermore, it is observable that both papers mentioned that it had used K-Nearest Neighbour and Logistic Regression as the algorithm for the ML model [27, 28, 29]. Thus, utilizing different algorithms provide insightful information on how processed data can be used to create a diagnosis model.

2.2.6 Study 7: Transformer Models in Healthcare: A Survey and Thematic Analysis of Potentials, Shortcomings and Risks

The rise of artificial intelligence in the field of medicine has slowly paved way to improve various aspects of medicine, ranging from services to quality of medicine. However, the abundance of data left unused is considered as wasted potential for artificial intelligence since existing studies highlight the ability of such in gathering essential and insightful forms of analysis. Likewise, the study titled, *"Transformer Models in Healthcare: A Survey and Thematic Analysis of Potentials, Shortcomings and Risks"* explores existing

transformer models that are made to improve the current healthcare system. Furthermore, the paper highlights multiple benefits of implementing a transformer model, mainly: (1) Efficiency between Clinical Tasks, (2) Enhanced Data Analysis, (3) Improved System Approaches to Diagnosis, etc. [42]. However, the paper also defines the risks pertaining to inaccuracy and over reliance to artificial intelligence, especially that improving AI systems are now capable of doing human-related tasks. Thus, the paper explains that in using a transformer model in the field of healthcare and medicine, such models are to be used as support systems and must involve forms of human intervention to both improve the quality and accessibility of such systems [42].

2.2.7 Study 8: A deep learning transformer model predicts high rates of undiagnosed rare disease in large electronic health systems

Within medical institutions, majority have shifted their documentation processes from paper-based to electronic-based records, called Electronic Health Records (EHR) [43]. Moreover, the over abundance of data fed to EHR has given the opportunity for these institutions to leverage the gathered data. These data, when studied and analyzed, is said to be a key in determining the possible factors of acquiring a distinct disease, specifically that these may improve both the healthcare quality and services provided [44, 43]. Using the Transformer Model was a crucial step in determining any underlying factors causing rare genetic diseases, specifically for patients flagged as undiagnosed [44].

Furthermore, it is emphasizes that creating a systematic framework that enacts the process of sorting and training is essential in determining which factors are needed to take into consideration for the prediction of rare genetic diseases between patients. Similarly, the development of the appendicitis prediction model may utilize the processes and methods taken in this research journal, especially that large datasets will be utilized to train the transformer model. Likewise, to determine the efficacy and efficiency of the model, logistic regression analysis was used to analyze the outputs produced by the model, implying that even with rigorous data training in the model, it is best to practice data output analysis for improvements and to properly conclude the analysis gained from the model [44].

2.2.8 Study 9: Learning the natural history of human disease with generative transformers

Pattern and data recognition are paramount features in detecting relationships and correlations between elements, variables, and data inputs. In the field of healthcare, each symptom, each medicine, and each medical intervention allocated to the provision of medical services are all essential components. Furthermore, the process of human disease and how it evolves into stages and levels are certain aspects of the medical field which are often considered a mystery, thus enabling experts to focus on finding out such. Health records and the progression of a disease between individuals is described as a fluctuating or linear trajectory with patients varying in cases and experiences [45]. Now, given that there are re-occurring patterns for disease (e.g. Flu, where a flu can happen to anyone at any time), using generative transformers or transformer models can create a great impact into determining plausible outcomes based on the aforementioned re-occurring patterns. Not only does the model become limited to fluctuating patterns but linearly continuous patterns are also determined based on aspects gradually affected and worsened over time.

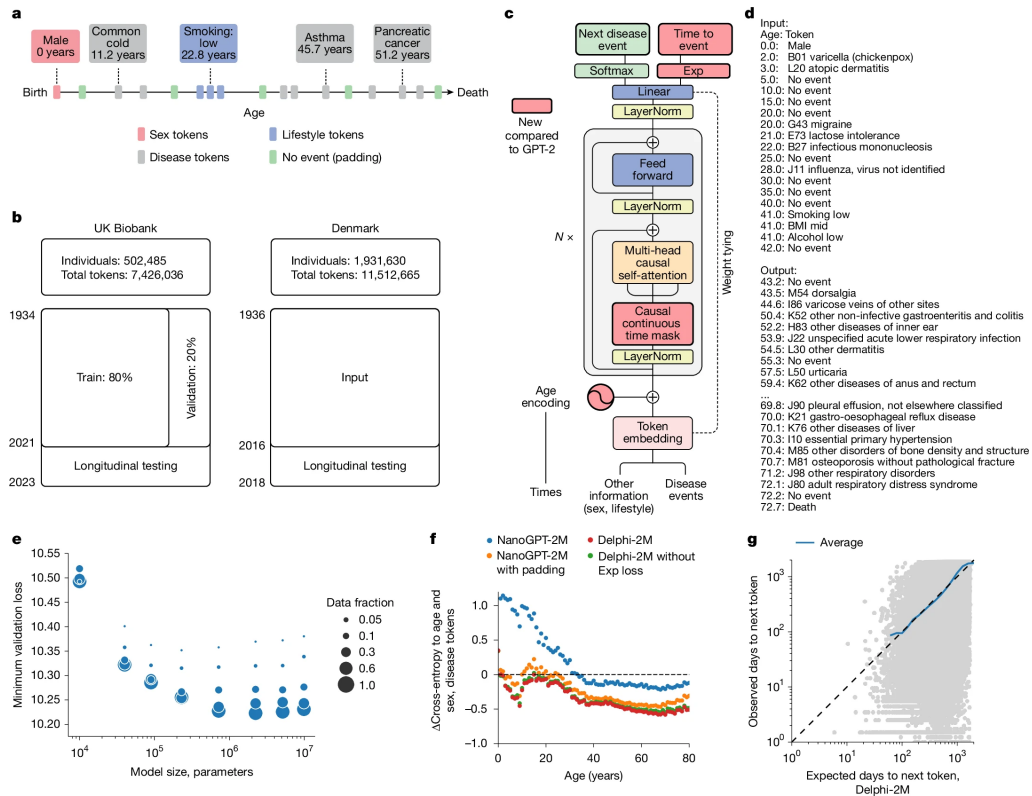


Figure 2. Delphi, a modified GPT architecture, models health trajectories

Source: Shmatko, A., Jung, A. W., Gaurav, K., Brunak, S., Mortensen, L. H., Birney, E., Fitzgerald, T., & Gerstung, M. (2025). Learning the natural history of human disease with generative transformers. <https://doi.org/10.1038/s41586-025-09529-3>

To create a comparison analysis between elements, assumed to be correlated, the paper utilized Delphi, a systematic framework or approach that analyzes and visualizes certain trends or the so-called "*trajectories*" [45]. Furthermore, majority of diseases are tackled on a singular basis, meaning that medical tools are only able to detect sicknesses for one disease and a new test run should be executed for another disease. This itself is a concern highlighted in the paper due to less attention is allocated to detecting all possible diseases [45]. More over, Delphi-2M is a transformer model [View Figure 3.] that cross-studies and correlates different symptoms, sicknesses, and diseases based on pre-existing and current datasets available. It is important to note that Delphi, a transformer model, applies tokenization to simplify sequences between data A to data B. The study and application of tokenization in the development of the diagnosis prediction model for pediatric appendicitis will be crucial in creating a hierarchical path from one variable/parameter to another, while these tokens act as nodes, repetitive training will improve the models ability to detect possible correlations between these variables.

On the other hand, the paper mentions that Delphi uses a continuous age encoding for these the diseases, implying that time play a crucial role when detecting diseases and when concluding a diagnosis [45]. Given that it requires data samples that range from a certain time period to fully recognize the correlation of each element, its application will be harder to implement for acute pediatric appendicitis since acute appendicitis happen from a timespan of hours to days [6, 7, 3, 45]. Nonetheless, the development of the appendicitis diagnosis prediction model should factor in the time of symptom onset given a short amount of allotted time per patient.

2.3 Limitations and Benefits of Artificial Intelligence to Appendicitis Diagnosis

Provided the applications and benefits that artificial intelligence, there comes limitations which affects its application in the medical field. In the earlier parts of the paper, limitations regarding ethics and moral-related issues are discussed due to how AI may be used. In this case, the technical limitations of AI is the process of data collection and the use of the proper algorithm is present [46]. As mentioned in the research paper written by Lam, et al. 2023, they had used numerous algorithms for different purposes and for comparisons [27]. This posses a problem for medical professionals and health

informatics as deciding which algorithm suits the best is then challenged as it forces the person to test on all suitable and possible algorithms. Therefore, this research paper will utilize the use of transformer models to eliminate the need to overuse algorithms for various variables.

2.4 Conclusion

Artificial Intelligence, as a whole, encompasses a massive field of study that is paramount when applied in various fields of study, specifically in the field of medicine. Nonetheless, utilizing an AI model that has a self-attention mechanism in the development of diagnosis prediction model based on pre-existing datasets can open newer applications of the transformer model in the medical field. Moreover, the limitations of AI is crucial and will be taken into consideration knowing what the research can handle and what the research specifically tackles, provided with the current resources. Furthermore, pre-existing researches recognizes the possible and potential benefits that can be acquired from utilizing artificial intelligence and the transformer model in the development of the diagnosis prediction model for medical practices.

3 Theoretical Framework

3.1 Appendicitis Detection

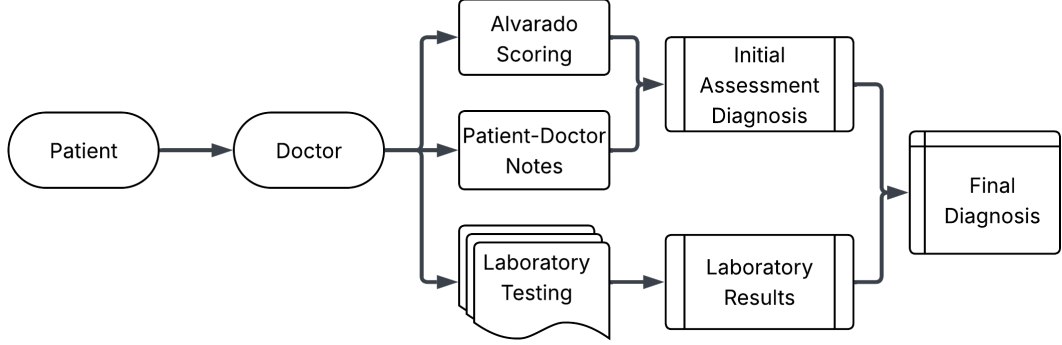


Figure 3. Appendicitis Diagnosis Process

In appendicitis detection, majority of cases start with patient-doctor consultations to determine various key factors that may cause appendicitis [View Figure 3.]. However, these data are categorized into different aspects to create a structured format for analysis and to limit any unnecessary data overflow. The six (6) categories are: Demographic Information, Clinical and Symptomatic Variables, Scoring Systems, Laboratory Findings, Ultrasound and Imaging Findings, and the Diagnostic and Outcome Variables.

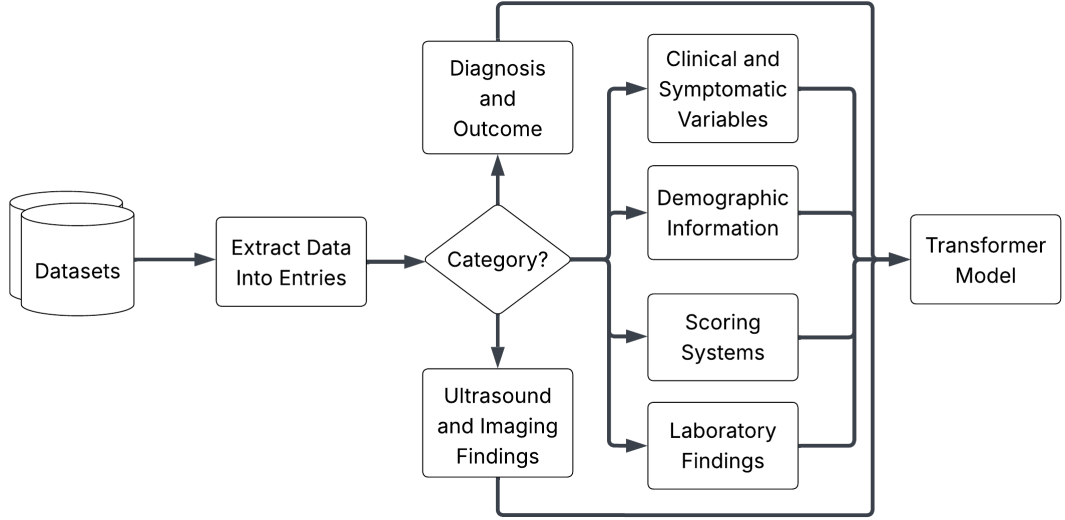


Figure 4. Dataset Sorting

To apply data categorization in the appendicitis diagnosis prediction model, segregated data must be taken into account to find correlation between elements [View Figure 4.]. Theoretically, developing the appendicitis diagnosis prediction model would require the use of the diagnosis accuracy theory, which focuses on determining the presence

of the disease based on current and pre-existing indicators. Given such, the categorized variables will be transformed into vectorized embeddings to be mapped out for the transformer model. Through data mapping between correlated variables, it is assumed that the transformer model can learn patterns that would simulate plausible clinical and medical reasoning while learning the most from currently available data.

3.2 Artificial Intelligence for Detection

Artificial Intelligence or AI is a focus under computer science that utilizes of digital technologies as means of performing activities and tasks often done by human beings, often encapsulating human-related skills and abilities [47, 48]. Various human-related skills and abilities include logical reasoning, pattern detection, data analysis, problem-solving, and overall, developing experience and knowledge. These improving skills within AI systems, when combined and connected amongst one another, aids in creating a systematic process often seen in highly intelligent beings. Moreover, such systematic processes can be used in the field of detection such as text detection, language detection (e.g. Google Translate), image detection (e.g. Google Image Search), etc. It is worth noting that neural networks and transformer models are primary techniques used to develop these aforementioned detection systems.

Neural Networks are machine learning models that are focused on tasks that are related to pattern and data recognition with data can infinitely span to achieve a better trained model [30, 49]. Moreover, the Transformer Model contains a self-attention mechanism which has a primary goal of allocating its memory and focus towards data inputs that deem to be crucial in the overall analysis of the data [31].

Provided that transformer models and neural networks aid in creating a framework of connections and relationships between factors, variables, and elements, the applications of AI detection can range from different fields of expertise and studies. Research papers such as *"Evaluating the efficacy of AI content detection tools in differentiating between human and AI-generated text"* by Elkhatat, A. (2023) and *"Artificial Intelligence Tools for Detection, Research and Writing"* by Quinn, B. (n.d.) are papers that emphasizes the effectiveness of AI's ability to detect patterns and data based on its ability and knowledge of the field of study [50, 51]. Hence, it opens a path for a more in-depth application towards data detection for medical-related studies specifically disease detection.

3.3 Diagnostic Accuracy Theory

A theory that is often highlighted in medical and clinical diagnosis is the diagnostic accuracy theory. This refers to the ability of a diagnostic test to be able to fully identify all plausible diseases given related information [52, 53]. In the paper titled *"Diagnostic Testing Accuracy: Sensitivity, Specificity, Predictive Values and Likelihood Ratios"* by Shreffler, J. and Huecker, M. 2023, the theory behind diagnostic testing accuracy was discussed and dissected into four (4) different metrics, namely: Sensitivity, Specificity, Predictive Values, and Likelihood Ratios [53]. The aforementioned metrics are all dependent on one another, provided that each metric is focused on testing the accuracy in the four (4) fields.

Provided with four (4) different metrics, it is important to note that various diseases will also have varying values, sensitivities, specificities, and likelihood values. This itself emphasizes that various utilized diagnostic tools must be weighted and differentiated when approaching diseases, since it should be expected that similar symptoms do not generally equate to the same singular disease. Now, with the weight of accuracy and efficiency being balanced, the paper theorized that heavily focusing on certain metrics will counter the increase of other metrics, especially that for accuracy to be fully attained, longer time is spent waiting for results and evidences to prove a diagnosis. Whereas, the efficiency of each tool in detecting a disease may vary given circumstances ranging from medical quality and time spent. The theory simply defines that to focus on accuracy means one must devote more time and resources to attain a higher accuracy rate within the four (4) metrics and to focus on efficiency means that the validity of each metric is to be questioned [53, 54].

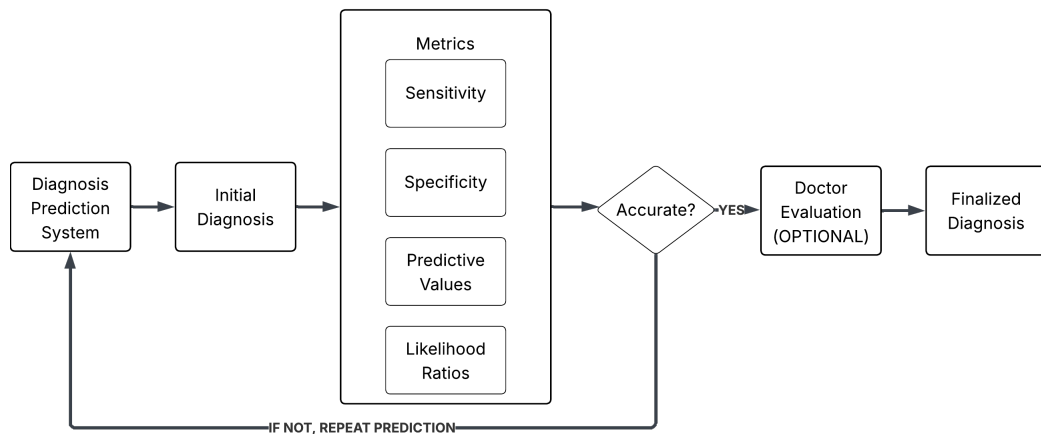


Figure 5. Application of Diagnostic Accuracy Theory

Consequently, in the development of the appendicitis diagnosis prediction model, prior paragraphs mentions how accuracy is paramount in providing a successful diagnosis. Thus, Figure 3. is a flow of actions, assuming that the diagnosis accuracy theory is to be implemented in the model. At the first and initial diagnosis, the output values are to be measured using the aforementioned four (4) metrics to determine whether it is accurate or not. If accuracy rates are lower than 60% then another run must be done in the diagnosis prediction model since accuracy rates lower than 60% is unsafe and unreliable. Likewise, if accuracy rates are equal or higher than 60% then the outputs are to undergo the Doctor's evaluation before finalizing a diagnosis.

3.4 Representation Learning Theory

Representation learning theory is a foundation built to focus on a multifaceted problem related to machine learning systems and algorithms. The theory is said to be paramount in dealing with processes involved with learning relationships between features that captures the structure of the hierarchical patterns, often highlighting crucial features [55]. As artificial intelligence evolves into a multi-level paradigm, representation learning become a core fundamental aspect in pattern detection and data recognition.

In a paper titled *"Representation Learning: A Review and New Perspectives"* by Bengio, Y., et al 2013, mentioned that *"Transfer learning is the ability of a learning algorithm to exploit commonalities between different learning tasks in order to share statistical strength, and transfer knowledge across tasks"* [56]. Although, transfer learning is not the main focus of the appendicitis diagnosis prediction model, this itself emphasizes how data entries from datasets may be correlated with one another and with proper data fine tuning [using Transformer Model's Self-Attention Feature], the model's focus will be shifted to what may be deemed as crucial and essential in predicting a diagnosis. It is important to note that Representation Learning Theory stresses that for data to be fully utilized in the field of any study, data must be extracted and embedded in a deep learning or artificial intelligence model for data analysis and data visualization.

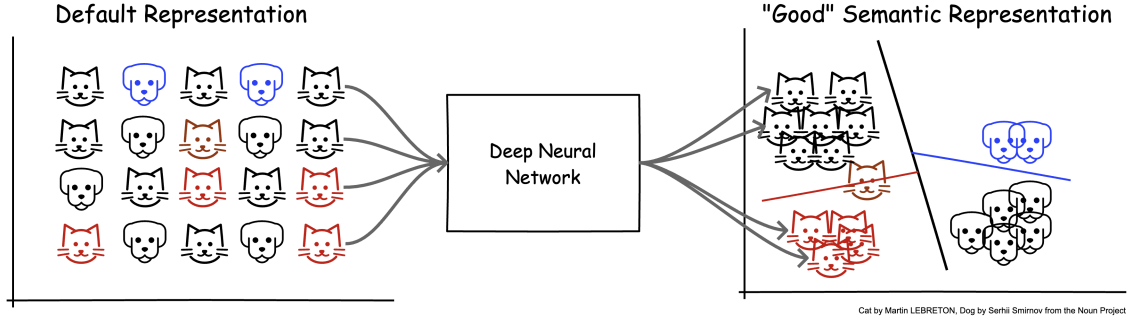


Figure 6. Representation Learning Example

Source: Representation Learning 101 for Software Engineers. (2020). Fastforwardlabs.com.
<https://blog.fastforwardlabs.com/2020/11/15/representation-learning-101-for-software-engineers.html>

As transformer models are examples of structured deep learning neural networks, being able to visually see data sorting and segregation is crucial in determining correlated samples and entries whilst accounting for outliers and shared relationships. In Figure 4., it is a diagram depicting how unsorted data, when fed into a deep neural network (DNN), can be sorted into categories. Further, such sorting practices can be used in the appendicitis detection model due to the vast categories between data, ranging from medical laboratory results to basic demographic data. Of course, such data cannot be sorted as is since varying values mean, more segregation, thus harder to find correlation. So, in the diagnosis model, data will be fed and analyzed based on data categories to ensure the integrity of the data and to firmly establish connections without creating unnecessary extra sorting between data entries.

Deep learning methods and practices, specifically transformer models, practice such systematic frameworks by learning data entries in a hierarchical setting [45, 56]. From data pre-processing techniques, these transformers read through datasets, converting each data entries to vector embeddings whilst emphasizing the relationship between variables [30, 49, 34, 45]. Similarly, data entries originating from tabular medical and patient data (e.g. Symptoms, Laboratory Results, etc.) can be embedded to visualize patterns and relationships between contents [7, 57, 34, 45].

3.5 Transformer Models

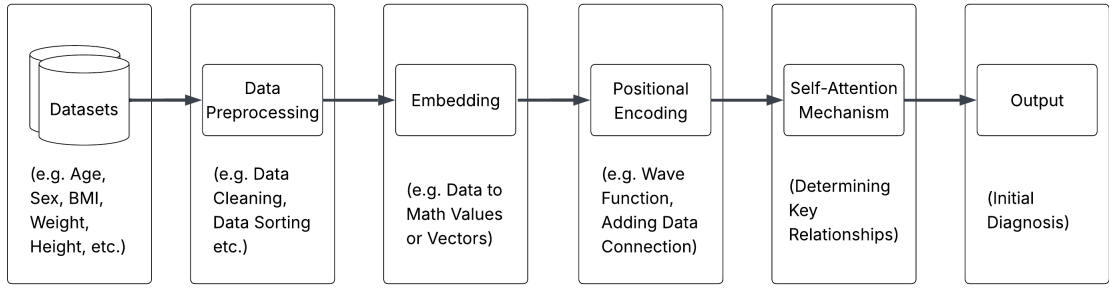


Figure 7. General Process For A Transformer Model

Embeddings, specifically under Transformer Models, can be defined as the process of converting categorized and sorted data into vector representations to visualize and determine the key factors and relationships between elements. Do note that embeddings are crucial in artificial intelligence or computer systems due to the fact that computer systems understand data in the form of numerical and mathematical computations rather than the words that we understand. Through conversion to mathematical values, overlaying the said data is paramount in the layering data process of a transformer model.

In deep learning algorithms, data training is said to be analyzed in a non-sequential pattern or a non-linear movement. However, in the case of appendicitis detection, data such as laboratory results cannot go first than your patient-doctor consultations, thus, implying that linear or sequential processes are crucial in determining the step-by-step process in diagnosis appendicitis. Therefore, positional encoding is the process of applying wave functions to mathematical computations (vectorized embeddings) to ensure that relationships between data are consistent throughout the data training process.

Self-Attention is a specialized feature utilized by transformer models to analyze and determine the relationships between all present variables considering a sequence within the data entry. For majority of older applications of neural networks which focus on processing data one at a time, transformer models alongside the self-attention feature allows the model to simultaneously determine the most important or has the most important role between variables. Subsequently, the ability of the system to determine key relationships show signs of heightened awareness to important variables.

3.6 Decision Trees, Gradient Boosting, and XGBoost

In the processes of determining the efficiency of a transformer model in the field of diagnosis prediction for appendicitis, the usage of Decision Trees, XGBoost, and Gradient Boosting are utilized as means of comparison using the same dataset.

Decision Trees is a machine learning algorithm that is commonly used for classification and regression-related tasking due to its tree-like structure that allows data to be ordered in a sequential pattern [58]. Compared to various algorithms and models, decision trees are considered as the easiest to comprehend as it can be visualized as a tree with branching options. Moreover, decision trees perform a selection process called feature selection, a process that reduces inputs by gathering all necessary and important features, to determine different case scenarios or options. To add, a root node is the starting point for all cases and leaf/child nodes are sub-nodes of the root node which can be considered as a subset of the root.

Gradient Boosting is a process of creating and building prediction models sequentially, such that it focuses on combining predictive outputs from models considered as weak learners like Decision Trees. Subsequently, outputs from a single model will be analyzed and will be redone while taking into consideration all of the errors produced by the previous output [59, 60]. Therefore, Gradient Boosting is a repetitive machine learning process that will continuously retry and redo its learning processes by utilizing decision trees while eliminating any newly generated errors for a better and more accurate output.

Similarly, XGBoost or eXtreme Gradient Boosting is considered as a stronger and more advanced version of the Gradient Boosting, while it uses the same framework as Gradient Boosting, the overall processes done for XGBoost is far more optimized but may require a stronger processor to processes any form of machine learning and artificial intelligence techniques.

The idea of choosing Decision Trees, Gradient Boosting, and XGBoost is made as means expanding the applications of decision trees due to its ability of detecting sequential patterns through branching out variables, as well as the fact that Gradient Boosting and XGBoost will utilize Decision Trees as a foundation for the model. All three (3) models mentioned will be used as comparison for the Transformer Models in terms of determining the efficiency of such model in the field of diagnosis prediction.

4 Design and Implementation

4.1 System Design

The proposed system plans to predict a diagnosis for pediatric appendicitis patients using artificial intelligence transformer models. It is envisioned that the said model will contain multiple core features which are dependent in producing an accurate diagnosis. Patient data is the base component of the system as data such as: demographics, symptoms, laboratory results, etc. are to be extracted from these datasets. The extracted data will undergo a series of pre-processing techniques to eliminate inconsistencies within the data, whilst maintaining the integrity of the original dataset. After thorough data cleaning, the data is transformed to vector embeddings or representations suitable for the AI model. Provided vectorized data entries, the model is assumed to analyze, detect, and highlight the major relationships between each variable. More over, the self-attention feature of the transformer model aims to capture how each variable is dependent (e.g. how symptoms are dependent to laboratory results). Consequently, the model outputs a diagnosis prediction provided with evidences to support the prediction. These predictions will be measured with different metrics to gauge the accuracy of the produced diagnosis outputs.

4.2 Input and Output Specifications

4.2.1 Input-Related Specifications

Given the datasets available for research purposes, the study will opt to utilize the variables used in the detection of appendicitis. More so, the following data inputs are required to maintain a unified flow to eliminate data inconsistencies. Importantly, it should be mentioned that:

1. Fields related to Gynecology will be separated from men and women, especially that different genders may generally have different outcomes while same genders may produce like outcomes.

Category	Features / Variables	Data Type(s)
Demographic Information	Age, Sex, Height, Weight, BMI, Length_of_Stay	Numerical, Categorical
Clinical and Symptomatic Variables	Migratory_Pain, Lower_Right_Abd_Pain, Contralateral_Rebound_Tenderness, Ipsilateral_Rebound_Tenderness, Coughing_Pain, Nausea, Loss_of_Appetite, Dysuria, Peritonitis, Psoas_Sign, Stool, Severity, Management	Binary, Categorical
Scoring Systems	Alvarado_Score, Paediatric_Appendicitis_Score	Numerical
Laboratory Findings	Body_Temperature, WBC_Count, Neutrophil_Percentage, Segmented_Neutrophils, Neutrophilia, RBC_Count, Hemoglobin, RDW, Thrombocyte_Count, Ketones_in_Urine, RBC_in_Urine, WBC_in_Urine, CRP	Numerical, Binary
Ultrasound and Imaging Findings	Appendix_on_US, Appendix_Diameter, Appendix_Wall_Layers, Appendicolith, Free_Fluids, Target_Sign, Perfusion, Perforation, Surrounding_Tissue_Reaction, Appendicular_Abscess, Abscess_Location, Pathological_Lymph_Nodes, Lymph_Nodes_Location, Bowel_Wall_Thickening, Conglomerate_of_Bowel_Loops, Ileus, Coprosta-sis, Meteorism, Enteritis, Gynecologi-cal_Findings, US_Performed, US_Number	Binary, Cate-gorical, Numerical
Diagnostic and Outcome Variables	Diagnosis, Diagnosis_Presumptive	Categorical

4.2.2 Output-Related Specifications

Assumed that the data used in the transformer model is aligned accordingly for both variables and features, the model is presumed to determine and predict a plausible diagnosis given an input of data. Likewise, a confidence score and severity prediction will be implemented to determine the probability of a patient experience appendicitis given their case and if so, how severe may their case be. Note that a "*severity_prediction*" output will be utilized in cases where data is deemed inconsistent and lacking within the datasets provided that less data, affects the model to accurately predict a diagnosis.

Data Output	Description	Data Type(s)
Diagnosis_Prediction	The final predicted diagnosis using the trained transformer model	Categorical (Appendicitis or No Appendicitis)
Confidence_Score	Probability that given the diagnosis prediction is correct and accurate	Float (0 - 1)

4.3 System Architecture

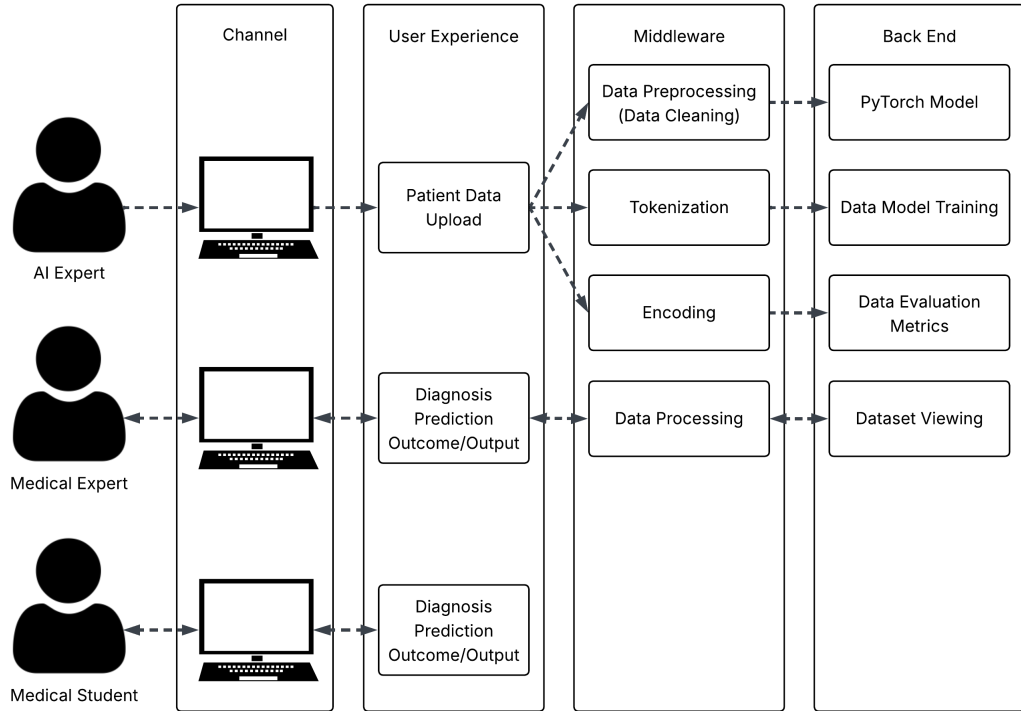


Figure 8. System Architecture

The system architecture assumes that in the development of the diagnosis prediction

mode, three (3) roles will be made, namely: the AI Expert, Medical Student, and Medical Expert.

The AI Expert's role is to update the system by uploading patient datasets which will undergo a series of Data Preprocessing, Tokenization, and Encoding. For each aforementioned series of processes, it will then be used in the PyTorch Model (Transformer Model), Data Model Training, and Data Evaluation Metrics (Sensitivity and Accuracy).

The Medical Student and Medical Expert's Primary Role is to use the system to predict a diagnosis. For the Medical Expert's Specific Roles, data viewing is a feature that will allow the user to view details regarding a patient, but it has to undergo data processes to fully display all essential data and information back to the Medical Expert's device. Note that produced diagnosis prediction must be reviewed to determine if the produced output is accurate or correct.

4.4 System Diagrams

4.4.1 Pipeline Diagram

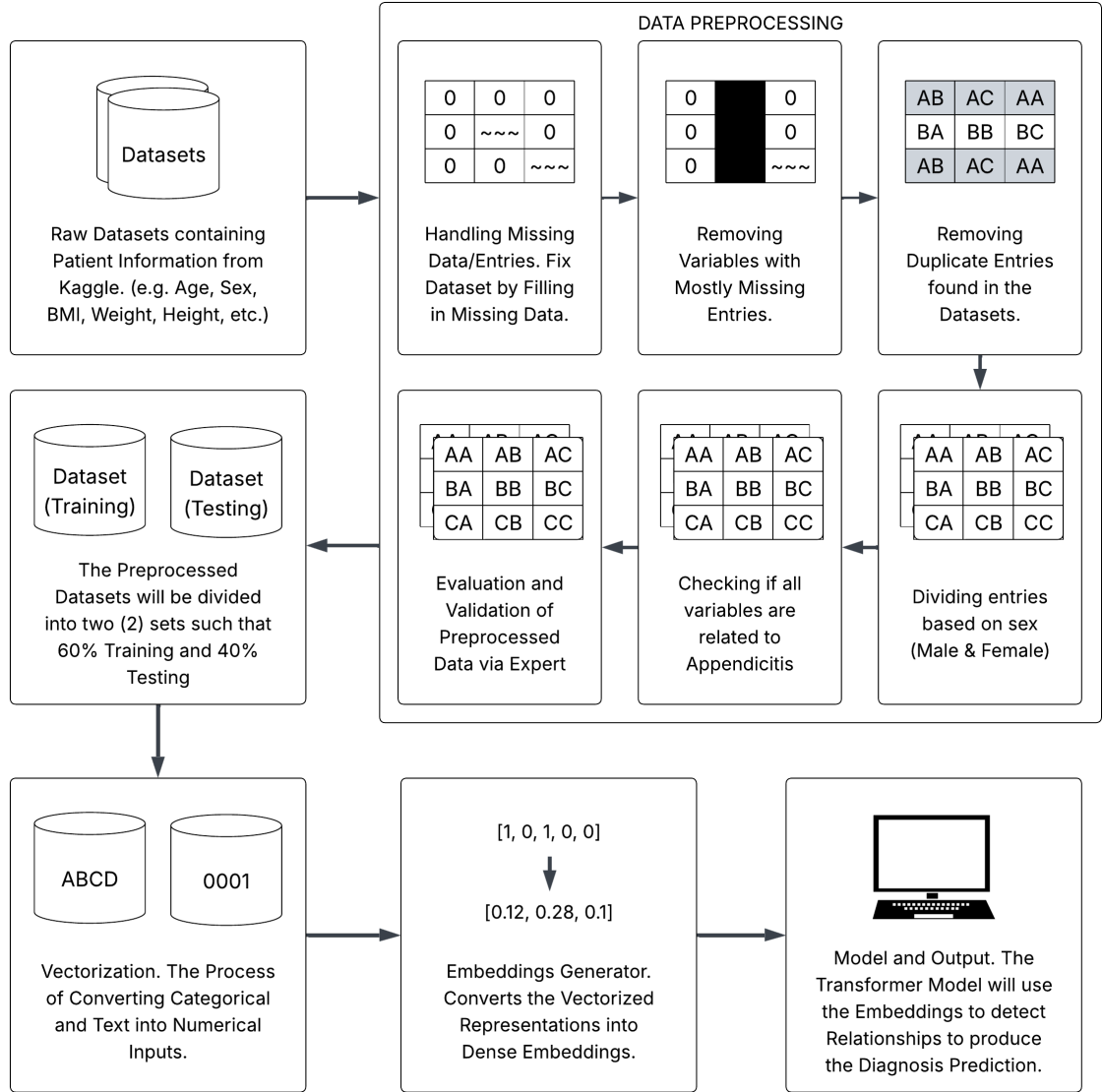


Figure 9. Pipeline

The Pipeline Diagram shows the development process of the diagnosis prediction model. The flow starts with the preparation of the datasets coming from Kaggle which is to undergo a series of preprocessing techniques namely: Handling Entries, Removing Variables, Removing Duplicates, Dividing Entries, Rechecking and Evaluation of Preprocessed Datasets. After undergoing data preprocessing, the datasets are partitioned with a 60:40 ratio for training and testing, respectively. The 60% partition dataset will be vectorized for the embeddings generator to develop the transformer model. Consequently, after training, the 40% partition will be used to test and validate the system.

4.4.2 Data Flow Diagram

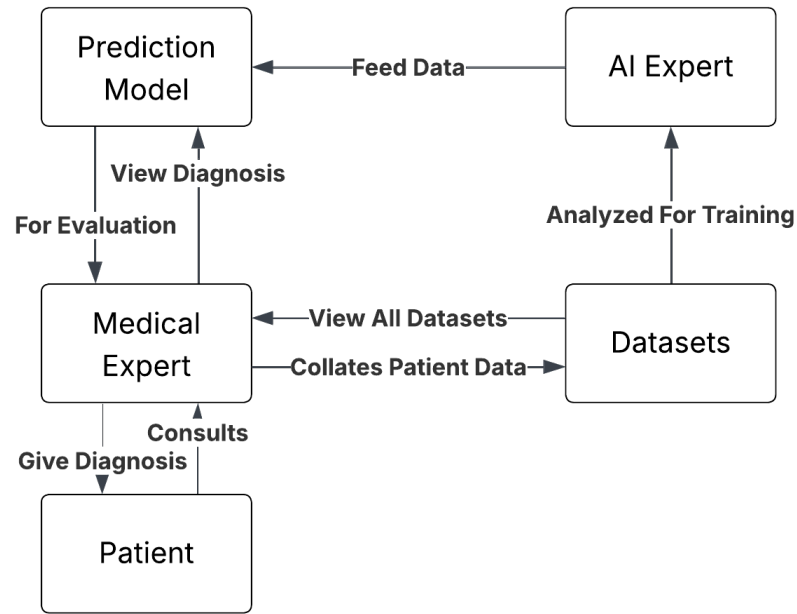


Figure 10. Data Flow Diagram (DFD)

The Data Flow Diagram demonstrates the flow of actions from user to system and vice versa. Data is extracted from the patient and would be collated by the expert to undergo data preprocessing techniques. The AI feeds the data to the prediction model which produces the diagnosis to be stored in the data storage. Do note that a medical student may have the option to use the diagnosis prediction model, however the data flow diagram highlights the interaction between Medical Expert and the system managed by the AI Expert.

4.4.3 Use Case Diagram

The figures below are the use case diagrams of the diagnosis prediction AI model. The three (3) main users are the Medical Student, Medical Expert, and the AI Expert. The Medical Student and Medical Expert are the primary user of the prediction model with capabilities ranging from creating predictions to adding new patient data for analysis and diagnosis prediction. The AI Expert is focused on managing the system to cater to the needs of the medical expert in terms of datasets, metric evaluations, and definition of variables and attributes.

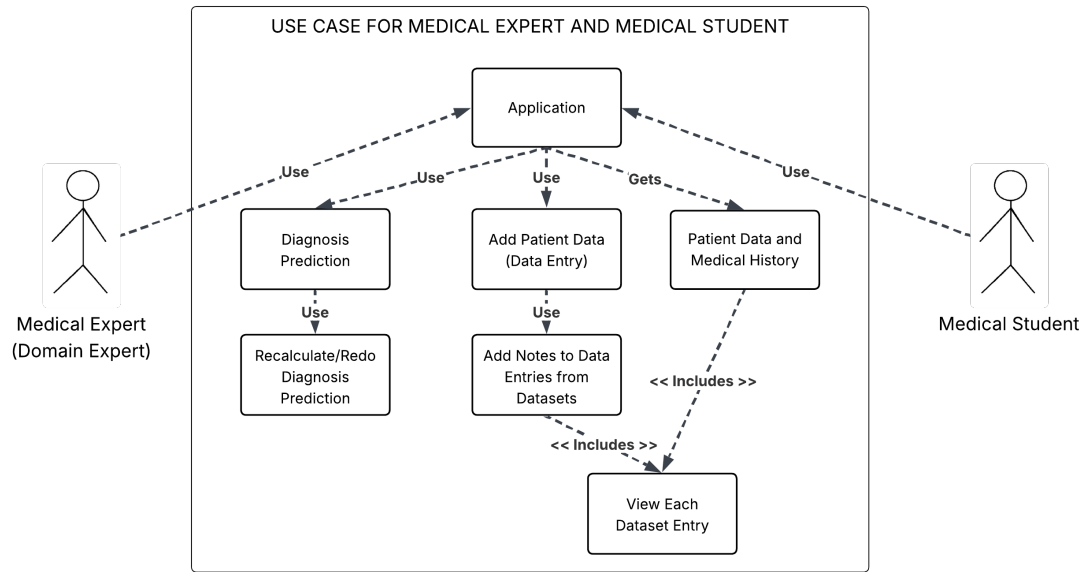


Figure 11. Medical Expert Use Case Diagram

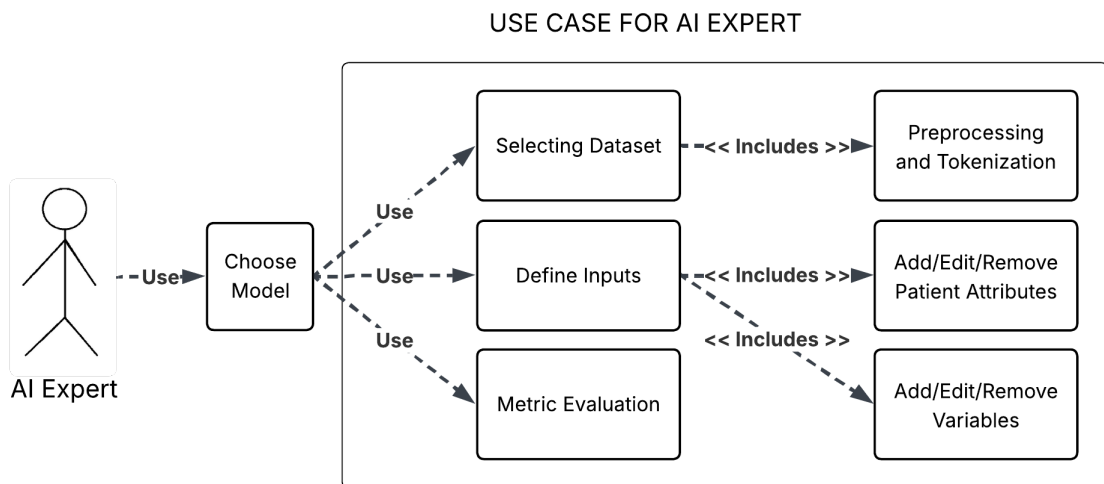


Figure 12. AI Expert Use Case Diagram

4.5 Software Requirements

In the development of the Appendicitis Diagnosis Prediction Model, the project requires the following software and tools to aid in its creation. Further, alternative tools may be used along the process of development if the current mentioned software and tools are complex and difficult to use given the dataset and researcher's skill.

Software	Description
Python	The main programming language to be used in the development of the appendicitis diagnosis prediction model.
PyTorch	Deep Learning Framework to develop and train the Artificial Intelligence Transformer Model
Pandas and NumPy	Data Preprocessing and Data Cleaning to remove inconsistencies and possible skewness found within the datasets.
Scikit-learn	Tools to aid in the evaluation metrics.
Matplotlib	Visualization tools for dataset analysis.
SQL	Source of data extraction from Kaggle Datasets (Datasets are in .xlsx and .csv formats).

4.6 Evaluation Metrics

After the initial development of the appendicitis diagnosis prediction model, it is said that systems need to be evaluated based on certain metrics and criteria to determine the strengths and weaknesses of the such. However, given that the produced product is a clinical diagnosis model, a wider scope for evaluation must be used to determine the following: (a) Accuracy, (b) Sensitivity, (c) True Positive to False Positive Ratio, etc. Metrics and Criteria are essential aspects to the development and revision of the model. Subsequently, a domain expert is to test the efficacy and accuracy of the model to determine which aspects need improvement to cater to the needs in a real-life medical settings.

Furthermore, the paper will utilize statistical computing tools to determine an estimate value of how efficient and accurate the model is in predicting a correct diagnosis. The statistical tools or equations are:

Metric	Definition / Description	Formula
Accuracy	Computation to gauge the overall ability of the model to correctly predict, sort, and classify the Appendicitis Cases given model training using datasets.	$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$
Precision	Computation to determine the count of correctly predictive Positive Cases amongst the used datasets. Tests the reliability of the model in producing proper diagnosis.	$\text{Precision} = \frac{TP}{TP + FP}$
Sensitivity	Computation to determine if the model is able to correctly identify True Positive (TP) Appendicitis cases amongst the used datasets.	$\text{Sensitivity} = \frac{TP}{TP + FN}$

Legend : TP = True Positive; TN = True Negative; FP = False Positive; FN = False Negative

4.7 Expected Output/Timeline

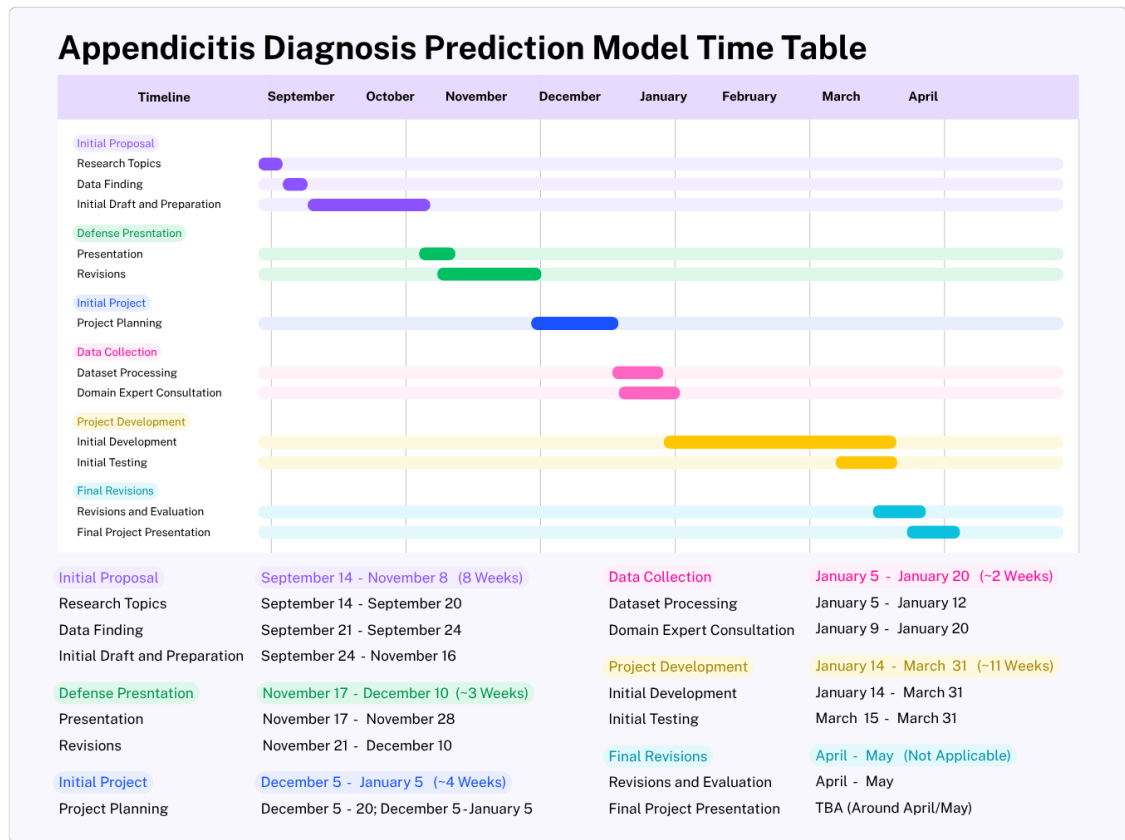


Figure 13. Gantt Chart

The Gantt Chart as shown from the figure above is a general assumption of the possible time table containing the list of activities to be done per week and per month. The activities are sorted based on the highlighted subsections. The total number of months that will take to accomplish the overall project is approximately eight (8) months.

References

- [1] M. Waseem and C. F. Wang, “Pediatric appendicitis,” in *StatPearls*. Treasure Island (FL): StatPearls Publishing, Jan. 2025.
- [2] S. Lotfollahzadeh, R. A. Lopez, and J. G. Deppen, “Appendicitis,” in *StatPearls*. Treasure Island (FL): StatPearls Publishing, Jan. 2025.
- [3] P. Ansari, “Appendicitis,” <https://www.msdmanuals.com/professional/gastrointestinal-disorders/acute-abdomen-and-surgical-gastroenterology/appendicitis>, Jul. 2024, accessed: 2025-9-20.
- [4] “Appendicitis,” <https://www.betterhealth.vic.gov.au/health/conditionsandtreatments/appendicitis>, accessed: 2025-9-20.
- [5] “The emergency condition called appendicitis,” <https://nnc.gov.ph/mindanao-region/the-emergency-condition-called-appendicitis>, Jan. 2023, accessed: 2025-11-14.
- [6] “Appendicitis test,” <https://my.clevelandclinic.org/health/diagnostics/22801-appendicitis-test>, Apr. 2022, accessed: 2025-9-22.
- [7] “Ruptured appendix: What does it mean?” <https://www.makatimed.net.ph/blogs/ruptured-appendix-what-does-it-mean>, Mar. 2022, accessed: 2025-9-22.
- [8] “Diagnosis of appendicitis,” <https://www.niddk.nih.gov/health-information/digestive-diseases/appendicitis/diagnosis>, Aug. 2025, accessed: 2025-9-21.
- [9] “CT scan versus MRI versus x-ray: What type of imaging do I need?” <https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/ct-vs-mri-vs-xray>, Aug. 2025, accessed: 2025-9-22.
- [10] A. A. Ismail, “When laboratory tests can mislead even when they appear plausible,” *Clin. Med.*, vol. 17, no. 4, pp. 329–332, Jul. 2017.
- [11] Y. Shen, F. E. Shamout, J. R. Oliver, J. Witowski, K. Kannan, J. Park, N. Wu, C. Huddleston, S. Wolfson, A. Millet, R. Ehrenpreis, D. Awal, C. Tyma, N. Samreen, Y. Gao, C. Chhor, S. Gandhi, C. Lee, S. Kumari-Subaiya, C. Leonard, R. Mohammed, C. Moczulski, J. Altabet, J. Babb, A. Lewin, B. Reig, L. Moy, L. Heacock,

- and K. J. Geras, “Artificial intelligence system reduces false-positive findings in the interpretation of breast ultrasound exams,” *Nat. Commun.*, vol. 12, no. 1, p. 5645, Sep. 2021.
- [12] O. O. Olateju, S. U. Okon, U. T. I. Igwenagu, A. A. Salami, T. O. Oladoyinbo, and O. O. Olaniyi, “Combating the challenges of false positives in ai-driven anomaly detection systems and enhancing data security in the cloud,” *Asian J. Res. Com. Sci.*, vol. 17, no. 6, pp. 264–292, Jun. 2024.
- [13] U. Mahmood, M. Cadogan, and U. M. A. Cadogan, “Alvarado score,” <https://litfl.com/alvarado-score>, Feb. 2019, accessed: 2025-9-22.
- [14] S. Shams Ud Din, I. Ullah Baig, M. Tassawar Hussain, A. Sadiq, T. Humayun, U. Ahmad, and A. Syed, “RIPASA versus alvarado score in the assessment of acute appendicitis: A prospective study,” *Turk. J. Surg.*, vol. 39, no. 3, pp. 231–236, Sep. 2023.
- [15] A. A. Abdullah, F. M. Katoof, and H. A. Jasim, “RIPASA versus modified alvarado scoring system in the diagnosis of acute appendicitis, as a new scoring system of acute appendicitis in ASIA,” *Int. J. Surg. Open*, vol. 15, pp. 42–45, 2018.
- [16] C. F. Chong, A. Thien, A. J. Mackie, A. S. Tin, S. Tripathi, M. A. Ahmad, L. T. Tan, S. H. Ang, and P. U. Telisinghe, “Comparison of RIPASA and alvarado scores for the diagnosis of acute appendicitis,” *Singapore Med. J.*, vol. 52, no. 5, pp. 340–345, May 2011.
- [17] M. Al-Tarakji, A. Zarour, R. Singh, and M. S. Ghali, “The role of alvarado score in predicting acute appendicitis and its severity in correlation to histopathology: A retrospective study in a qatar population,” *Cureus*, Jul. 2022.
- [18] S. Shams Ud Din, I. Ullah Baig, M. Tassawar Hussain, A. Sadiq, T. Humayun, U. Ahmad, and A. Syed, “RIPASA versus alvarado score in the assessment of acute appendicitis: A prospective study,” *Turk. J. Surg.*, vol. 39, no. 3, pp. 231–236, Sep. 2023.
- [19] Copeland and B. J., *artificial intelligence*, Sep. 2025.

- [20] C. Stryker and E. Kavlakoglu, “What is artificial intelligence (AI)?” <https://www.ibm.com/think/topics/artificial-intelligence>, Sep. 2025, accessed: 2025-9-22.
- [21] <https://www.unesco.org/en/artificial-intelligence>, accessed: 2025-9-22.
- [22] D. Cordero, Jr, “The downsides of artificial intelligence in healthcare,” *Korean J. Pain*, vol. 37, no. 1, pp. 87–88, Jan. 2024.
- [23] S. A. Alowais, S. S. Alghamdi, N. Alsuhebany, T. Alqahtani, A. I. Alshaya, S. N. Al-mohareb, A. Aldairem, M. Alrashed, K. Bin Saleh, H. A. Badreldin, M. S. Al Yami, S. Al Harbi, and A. M. Albekairy, “Revolutionizing healthcare: the role of artificial intelligence in clinical practice,” *BMC Med. Educ.*, vol. 23, no. 1, Sep. 2023.
- [24] K. Basu, R. Sinha, A. Ong, and T. Basu, “Artificial intelligence: How is it changing medical sciences and its future?” *Indian J. Dermatol.*, vol. 65, no. 5, p. 365, 2020.
- [25] “How artificial intelligence is disrupting medicine and what it means for physicians,” <https://learn.hms.harvard.edu/insights/all-insights/how-artificial-intelligence-disrupting-medicine-and-what-it-means-physicians/>, accessed: 2025-9-22.
- [26] M. Issaiy, D. Zarei, and A. Saghazadeh, “Artificial intelligence and acute appendicitis: A systematic review of diagnostic and prognostic models,” *World J. Emerg. Surg.*, vol. 18, no. 1, p. 59, Dec. 2023.
- [27] A. Lam, E. Squires, S. Tan, N. J. Swen, A. Barilla, J. Kovoov, A. Gupta, S. Bacchi, and S. Khurana, “Artificial intelligence for predicting acute appendicitis: a systematic review,” *ANZ J. Surg.*, vol. 93, no. 9, pp. 2070–2078, Sep. 2023.
- [28] H. Yazici, O. Ugurlu, Y. Aygul, M. A. Ugur, Y. K. Sen, and M. Yildirim, “Predicting severity of acute appendicitis with machine learning methods: a simple and promising approach for clinicians,” *BMC Emerg. Med.*, vol. 24, no. 1, Jun. 2024.
- [29] A. Erman, J. Ferreira, W. A. Ashour, E. Guadagno, E. St-Louis, S. Emil, J. Cheung, and D. Poenaru, “Machine-learning-assisted preoperative prediction of pediatric appendicitis severity,” *J. Pediatr. Surg.*, vol. 60, no. 6, p. 162151, Jun. 2025.
- [30] F. Lee, “What is a neural network?” <https://www.ibm.com/think/topics/neural-networks>, Sep. 2025, accessed: 2025-9-29.

- [31] D. Bergmann and C. Stryker, “What is an attention mechanism?” <https://www.ibm.com/think/topics/attention-mechanism>, Apr. 2025, accessed: 2025-9-29.
- [32] M. M. Ahmed, A. M. Sayed, D. El Abd, S. Fares, M. S. M. Said, and E. Elsayed Sedik Ebrahim, “Gender difference in perceived symptoms and laboratory investigations in suspected and confirmed COVID-19 cases: A retrospective study,” *J. Prim. Care Community Health*, vol. 12, no. 21501327211039718, Jan. 2021.
- [33] D. Kim, S. A. Butterworth, and R. D. Goldman, “Chronic appendicitis in children,” *Can. Fam. Physician*, vol. 62, no. 6, pp. e304–5, Jun. 2016.
- [34] S. Nerella, S. Bandyopadhyay, J. Zhang, M. Contreras, S. Siegel, A. Bumin, B. Silva, J. Sena, B. Shickel, A. Bihorac, K. Khezeli, and P. Rashidi, “Transformers and large language models in healthcare: A review,” *Artif. Intell. Med.*, vol. 154, no. 102900, p. 102900, Aug. 2024.
- [35] S. T. X. Next, “Data ecosystem explained – key elements & benefits of modern data solutions [2025],” <https://www.stxnnext.com/blog/data-ecosystem-explained>, Jun. 2025, accessed: 2025-9-29.
- [36] S. Madan, M. Lentzen, J. Brandt, D. Rueckert, M. Hofmann-Apitius, and H. Fröhlich, “Transformer models in biomedicine,” *BMC Med. Inform. Decis. Mak.*, vol. 24, no. 1, Jul. 2024.
- [37] H.-W. Ting, J.-T. Wu, C.-L. Chan, S.-L. Lin, and M.-H. Chen, “Decision model for acute appendicitis treatment with decision tree technology—a modification of the alvarado scoring system,” *J. Chin. Med. Assoc.*, vol. 73, no. 8, pp. 401–406, Aug. 2010.
- [38] P. K. Yadalam, P. V. Thirukkumaran, P. M. Natarajan, and C. M. Ardila, “Light gradient boost tree classifier predictions on appendicitis with periodontal disease from biochemical and clinical parameters,” *Front. Oral Health*, vol. 5, p. 1462873, Sep. 2024.
- [39] “‘diagnosis’ vs. ‘prognosis’,” <https://www.merriam-webster.com/grammar/usage-of-diagnosis-and-prognosis-difference>, May 2020, accessed: 2025-9-22.

- [40] R. F. Wolff, K. G. M. Moons, R. D. Riley, P. F. Whiting, M. Westwood, G. S. Collins, J. B. Reitsma, J. Kleijnen, S. Mallett, and for the PROBAST Group†, “PROBAST: A tool to assess the risk of bias and applicability of prediction model studies,” *Ann. Intern. Med.*, vol. 170, no. 1, pp. 51–58, Jan. 2019.
- [41] IBM, “Machine learning,” <https://www.ibm.com/think/topics/machine-learning>, Sep. 2025, accessed: 2025-9-22.
- [42] K. Denecke, R. May, and O. Rivera-Romero, “Transformer models in healthcare: A survey and thematic analysis of potentials, shortcomings and risks,” *J. Med. Syst.*, vol. 48, no. 1, p. 23, Feb. 2024.
- [43] “Electronic health records,” <https://www.cms.gov/priorities/key-initiatives/e-health/records>, accessed: 2025-9-30.
- [44] D. M. Jordan, H. M. T. Vy, and R. Do, “A deep learning transformer model predicts high rates of undiagnosed rare disease in large electronic health systems,” Dec. 2023.
- [45] A. Shmatko, A. W. Jung, K. Gaurav, S. Brunak, L. H. Mortensen, E. Birney, T. Fitzgerald, and M. Gerstung, “Learning the natural history of human disease with generative transformers,” *Nature*, Sep. 2025.
- [46] B. Khan, H. Fatima, A. Qureshi, S. Kumar, A. Hanan, J. Hussain, and S. Abdullah, “Drawbacks of artificial intelligence and their potential solutions in the healthcare sector,” *Biomed. Mater. Devices*, vol. 1, no. 2, pp. 731–738, Sep. 2023.
- [47] Copeland and B. J., *artificial intelligence*, Nov. 2025.
- [48] C. Stryker and E. Kavlakoglu, “What is artificial intelligence (AI)?” <https://www.ibm.com/think/topics/artificial-intelligence>, Oct. 2025, accessed: 2025-11-4.
- [49] S.-H. Han, K. W. Kim, S. Kim, and Y. C. Youn, “Artificial neural network: Understanding the basic concepts without mathematics,” *Dement. Neurocognitive Disord.*, vol. 17, no. 3, p. 83, 2018.
- [50] A. M. Elkhatat, K. Elsaid, and S. Almeer, “Evaluating the efficacy of AI content detection tools in differentiating between human and AI-generated text,” *Int. J. Educ. Integr.*, vol. 19, no. 1, Sep. 2023.

- [51] B. Quinn, “Artificial intelligence tools for detection, research and writing,” <https://guides.library.ttu.edu/artificialintelligencetools/detection>, Jan. 2023, accessed: 2025-11-4.
- [52] “National cancer institute (.gov),” <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/diagnostic-test>, Feb. 2011, accessed: 2025-10-3.
- [53] J. Shreffler and M. R. Huecker, “Diagnostic testing accuracy: Sensitivity, specificity, predictive values and likelihood ratios,” in *StatPearls*. Treasure Island (FL): StatPearls Publishing, Jan. 2025.
- [54] M. Khalifa and M. Albadawy, “AI in diagnostic imaging: Revolutionising accuracy and efficiency,” *Comput. Methods Programs Biomed. Update*, vol. 5, no. 100146, p. 100146, 2024.
- [55] N. Klingler, “Representation learning: Uncovering data patterns easily,” <https://viso.ai/deep-learning/representation-learning>, accessed: 2025-10-4.
- [56] Y. Bengio, A. Courville, and P. Vincent, “Representation learning: A review and new perspectives,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1798–1828, Aug. 2013.
- [57] B. Khosravi, A. D. Weston, F. Nugen, J. P. Mickley, H. Maradit Kremers, C. C. Wyles, R. E. Carter, and M. J. Taunton, “Demystifying statistics and machine learning in analysis of structured tabular data,” *J. Arthroplasty*, vol. 38, no. 10, pp. 1943–1947, Oct. 2023.
- [58] S. Krishnan, “Machine learning for biomedical signal analysis,” in *Biomedical Signal Analysis for Connected Healthcare*. Elsevier, 2021, pp. 223–264.
- [59] P. Kothiyal, “GBM vs XGBoost: Key differences & comparison explained,” <https://talent500.com/blog/understanding-the-difference-between-gbm-vs-xgboost>, Apr. 2024, accessed: 2025-11-25.
- [60] B. Clark and F. Lee, “What is gradient boosting?” <https://www.ibm.com/think/topics/gradient-boosting>, Nov. 2025, accessed: 2025-11-25.